Joel Meyers

Contact Information

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Research Interests

Theoretical cosmology and fundamental physics; neutrinos, light relics, and the dark sector; CMB lensing and delensing; big bang nucleosynthesis; gravitational waves; machine learning

Research Experience

Southern Methodist University (SMU), Department of Physics
Associate Professor, 2024 - Present
Assistant Professor, 2018 - 2024

Canadian Institute for Theoretical Astrophysics (CITA)
University of Toronto, 2012 - 2018
Senior Research Associate, Theoretical Cosmology

Editorial Positions

Physical Review D
Associate Editor, 2026 - Present

Education

The University of Texas at Austin, Austin, Texas, USA
Ph.D. in Physics, August 2012
- Dissertation Topic: *Inflation: Connecting Theory to Observation*
- Advisor: Professor Steven Weinberg

University of Wisconsin, Madison, Wisconsin, USA
B.S. in Physics and Mathematics, May 2006

Collaborations

- **CMB-S4**, 2015 - present
  - Governing Board 2025 - 2026
  - Science Council Co-Chair 2020 - 2024
  - Executive Team 2020 - 2024
  - Light Relics Working Group Co-Lead 2018 - 2019
- **Simons Observatory**, 2016 - present
- **PICO**, 2016 - present
- **CMB-HD**, 2018 - present
- **CCAT-Prime**, 2016 - present
  - Rayleigh Scattering Working Group Deputy Lead 2020 - present

Research Funding

- National Science Foundation, Astronomy and Astrophysics Research Grant AST-2510926,
  New Opportunities with Weak Gravitational Lensing,
  PI (Co-PI A. van Engelen, ASU), 2025 - 2027
- Department of Energy, Office of High Energy Physics, Grant DE-SC0010129 (theory),
  Research at the Energy Frontier at Southern Methodist University,
  Co-PI, 2019 - present
- NASA Astrophysics Data Analysis Program, Grant 80NSSC24K0665,
  Searching for New Signals in the Cosmic Microwave Background,
  Co-I (SMU subaward; lead institution Arizona State University), 2024 - 2027

Honors and Awards

- Buchalter Cosmology Prize, Second Prize, 2024
- Robert S. Hyer Award for Undergraduate Research (Faculty Supervisor), 2020
- Beatrice and Vincent Tremaine Fellowship, 2016 - 2017

Postdocs Supervised

- **Victor Chan**, SMU, January 2024 - January 2025
  - Current: Echologics, Toronto
- **Joseph Ryan**, SMU, September 2021 - July 2024
  - Current: Instructor, University of South Carolina
- **Cynthia Trendafilova**, SMU, September 2019 - August 2022
  - Next: CAPS Fellow, UIUC
  - Current: Postdoc, Case Western Reserve University

Graduate Students Supervised

SMU

- **Joshua Perez**, Ph.D. Student, May 2024 - present
- **Alvin Leluc**, Ph.D. Student, May 2023 - present
  - Texas Section APS award for outstanding graduate oral presentation, 2024
- **Brandon Stevenson**, (Co-supervisor), Ph.D. Student, May 2020 - May 2026
  - Current: Innovative Signal Analysis, Dallas
- **Jasmine Liu**, (Co-supervisor), Ph.D. Student, August 2022 - August 2024
  - Current: Goldman Sachs
- **Antone Amalbert**, (Co-supervisor), Ph.D. Student, May 2023 - May 2024
- **Michael Litke**, (Co-supervisor), M.S. Student, August 2022 - August 2023
  - Next: Ph.D. program, University of Hawaii
- **Ishwita Saikia**, (Co-supervisor), Ph.D. Student, August 2022 - July 2023
- **Eric Guzman**, Ph.D. Student, October 2018 - May 2023
  - SMU Computational Science and Engineering Fellowship
  - Next: AI Engineer, Honeywell
  - Current: Data Scientist, Travelers Insurance

CITA (research supervisor)

- **Victor Chan**, Ph.D. Student, University of Toronto, Summer 2018 - Fall 2023
  - Next: Postdoc, SMU
- **Alex Laguë**, Ph.D. Student, University of Toronto, Summer 2018 - Fall 2019
  - Current: Postdoc, University of Pennsylvania
- **Selim Hotinli**, Ph.D. Student, Imperial College London, October 2016 - May 2020
  - Current: Postdoc, Perimeter Institute
- **Matthew Wilson**, M.S. Student, University of Toronto, January - August 2016
  - Next: Ph.D., University of Toronto, 2022

- **Derek Inman**, Ph.D. Student, CITA, September 2013 - August 2014
  - Current: Postdoc, RIKEN

Undergraduate Students Supervised

SMU

- **Christopher Cook**, May 2024 - May 2026
  - Goldwater Scholar 2025
  - Current: Ph.D. student (fellowship), UT Austin
- **Joshua Ange**, August 2022 - May 2025
  - Goldwater Scholar 2023
  - Churchill Scholar 2025
  - NSF GRFP 2025 and 2026
  - Current: Ph.D. student, Columbia
- **Noah Pearson**, May 2019 - August 2021
  - Robert S. Hyer Award 2020
  - Current: Ph.D. student, Montana State University

First-Year Research Experience (FYRE)

- **Nikita Karthik**, August 2026 - present
- **Aidan Premirelli**, August 2026 - present
- **Keillian Davis**, August 2025 - May 2026
- **Danna Alden**, August 2025 - May 2026

CITA

- **Connor Sheere**, Summer 2016 - Summer 2018
- **Brayden Mon**, Summer 2017
- **Harrison Winch**, Summer 2016
- **Vivian Britto**, Summer 2014
- **Shenglin Jing**, September 2012 - August 2013

Invited Talks

- Seminar / Colloquium: 37
- Conference / Workshop Plenary Talk: 41
- Selected Invited Talks:
  - University of South Carolina, Physics Colloquium, February 2026
  - Baylor University, Physics Colloquium, September 2025
  - *Collider, Dark Matter, and Neutrino Physics 2025*, Texas A&M University, May 2025
  - University of New Mexico, Physics Colloquium, May 2025
  - *Neutrinos in Physics and Astrophysics*, University of California Berkeley, January 2025
  - Perimeter Institute, Cosmology Seminar, December 2024
  - *Weinberg Inaugural Conference*, University of Texas at Austin, November 2024
  - *Texas Section APS Meeting*, Southern Methodist University, October 2024
  - *N3AS Seminar*, University of California, San Diego (Remote), September 2024
  - *Collider, Dark Matter, and Neutrino Physics 2024*, Texas A&M University, May 2024
  - University of Texas at Dallas, Physics Colloquium, April 2024
  - Canadian Institute for Theoretical Astrophysics, Seminar, March 2024
  - Perimeter Institute, Cosmology Seminar, January 2024
  - Stony Brook University, C.N. Yang Institute Seminar, November 2023
  - University of Texas at Arlington, Physics Colloquium, March 2023

- University of Illinois, CAPS Seminar, October 2022
- *Cosmological Probes of New Physics*, University of Maryland, September 2022
- *Collider, Dark Matter, and Neutrino Physics 2022*, Texas A&M University, May 2022
- AAS 240, CMB Special Session, June 2022
- APS April Meeting, Neutrino Mini-Symposium, April 2022
- *CosmoPalooza*, Virtual, January 2022
- Illinois Institute of Technology, Physics Colloquium, November 2020
- Michigan State University, High Energy Physics Seminar, November 2020
- *Topics in Cosmic Neutrino Physics*, Fermilab, October 2019
- Institute for Advanced Study (Princeton), Astrophysics Seminar, May 2019
- *nu Physics in the CMB*, University of California, San Diego, November 2018

SUMMER SCHOOL LECTURER

- 5th Santa Cruz School on Multi-Messenger Astrophysics, UC Santa Cruz, July 2026: *Cosmic Microwave Background – Observations and their Interpretation*
- International Neutrino Summer School 2023, Fermilab, August 2023: *Neutrino Cosmology*
- Michigan Cosmology Summer School 2023, University of Michigan, June 2023: *Fundamental Physics with the CMB*
- SLAC Summer Institute 2017, SLAC, August 2017: *Neutrinos and Light Relics in Cosmology*

WORKSHOPS ORGANIZED

- *Theoretical Astroparticle and Cosmology Symposium in Texas (TACOS)*, 2022–2026
  - Co-founder and organizer, with K. Boddy, J. Dent, A. Long, and L. Strigari
  - Annual meeting of Texas faculty, postdocs, and graduate students in theoretical particle physics, astrophysics, and cosmology; attendance has grown from about 75 to about 130
  - Rotating host: SMU (October 2022), Rice U. (October 2023), UT Austin (October 2024), Sam Houston State University (October 2025), Texas A&M University (October 2026)
- *CMB-S4 Spring 2024 Collaboration Meeting*, Virtual, March 2024
- *CMB-S4 Spring 2023 Collaboration Meeting*, Virtual, April 2023
- *CMB-S4 Summer 2022 Collaboration Meeting*, U. Chicago, August 2022
- *13th CMB-S4 Workshop: 2021 Summer Collaboration Meeting*, August 2021
- *11th CMB-S4 Workshop: Cosmology and Astrophysics in the Next Decade*, July 2020
- *CMB in HD*, Flatiron Institute, December 2018
- *Neutrinos and (G)astrophysics in Large-Scale Structure*, CITA, December 2016

OTHER PUBLICATIONS

- Instructor's Solution Manual for *Lectures on Quantum Mechanics* by Steven Weinberg, Cambridge University Press, 2015

TEACHING EXPERIENCE

Instructor

- **SMU Courses:**
  - Foundations of Modern Cosmology - PHYS 4368/6368: Spring 2019, Fall 2020, Spring 2022, Spring 2023, Spring 2024, Spring 2026
  - General Relativity - PHYS 4350/6350/7350: Spring 2025, Fall 2026
  - Foundations of Physics - PHYS 6160: Fall 2023, Fall 2024, Fall 2025
  - Quantum Field Theory II - PHYS 7315: Spring 2020

- Quantum Mechanics II - PHYS 6336: Spring 2021
- Modern Physics - PHYS 3305: Fall 2022
- General Physics - PHYS 1307: Fall 2019
- Astro-eXtraordinary - SCI 4301 (co-taught): Spring 2022

- **Course and Curriculum Development:**
  - New course: Foundations of Modern Cosmology (PHYS 4368/6368), first offered Spring 2019
  - New course: General Relativity (PHYS 7350), first offered Spring 2025; cross-listed undergraduate version (PHYS 4350/6350) added Fall 2026
  - Both new courses count toward the B.S. and Ph.D. specializations in astrophysics and cosmology
  - Graduate curriculum overhaul (Graduate Curriculum Task Force, 2020 - 2021): course requirements, specializations, degree plan
  - Laboratory component of PHYS 3306 (taught by other instructors)
- Scientific Computing - Symbolic Computing, May 2016 and May 2017, CITA

Service and Leadership

- **SMU Physics Department Committees:**
  - Undergraduate Recruitment Task Force: Chair, 2025 - 2026
  - Faculty Hiring Committee: Chair, 2022/23, 2025/26; Member, 2018/19, 2023/24, 2024/25, 2026/27
  - Department Steering Committee: Member, 2025 - present
  - Graduate Committee: Chair, 2026 - present; Member, 2018 - 2020, 2023 - 2026
  - Seminar Series: Co-organizer, 2022 - present
  - Undergraduate Committee: Member, 2020 - 2023
  - Graduate Curriculum Task Force: Member, Fall 2020 - Spring 2021
  - Astrophysics Discussion: Co-organizer, 2019 - present
  - Theoretical HEP Discussion: Co-organizer, 2018 - 2020
- **SMU University Service:**
  - Guild of Marshals: August 2020 - present
- **CMB-S4 Leadership:**
  - Governing Board: 2025 - 2026
  - Science Council Co-Chair: 2020 - 2024
  - Executive Team Member: 2020 - 2024
  - Light Relics Working Group Co-Lead: 2018 - 2019
  - Workshop Organizer: 2020 - 2024
- **CCAT-prime Leadership :**
  - Rayleigh Scattering Working Group Deputy Lead: 2020 - present
- **Simons Observatory:**
  - Mentorship Program, Mentor (two postdoctoral mentees): 2025 - present
- **CITA Committees:**
  - Cosmology Discussion: Co-organizer, 2012 - 2018
  - Blackboard Discussion: Co-organizer, 2013 - 2018
  - Postdoc Hiring Committee: Member, 2013 - 2018
  - Jamboree: Co-organizer, 2014 - 2015

- **Funding Review Panels:**
  - NASA
  - National Science Foundation
  - Department of Energy
  - European Research Council
  - US-Israel Binational Science Foundation
  - Royal Society
- **Textbook Reviews:**
  - Elsevier
  - Cambridge University Press
- **Journal Referee:**
  - Physical Review Letters
  - Physical Review D
  - Physical Review X
  - Physics of the Dark Universe
  - Journal of Cosmology and Astroparticle Physics
  - The Astrophysical Journal
  - Communications in Theoretical Physics

Outreach

- Lead organizer, SMU viewing event for the April 2024 total solar eclipse: thousands of spectators; lead-up talks, demonstrations, and public engagement with students and postdocs
- Public astronomical viewing nights with the SMU Society of Physics Students (telescopes operated by graduate students), growing out of the eclipse event
- Television, radio, podcast, and newspaper interviews

PUBLICATIONS

Note: Conventions for author ordering vary significantly within the field of cosmology. The typical practice for researchers with a background in high energy theory (including me) is to use alphabetical author lists by default. Exceptions are often made to boost visibility of students or junior researchers by placing them first in the list. In this section, author lists are alphabetical except those labeled with an asterisk (*).

1. J. Meyers,
Measuring the Imprints of the Cosmic Neutrino Background,
Invited review, Annual Review of Nuclear and Particle Science, submitted.
2. (*) C. Trendafilova, S. Raghunathan, B. Wallisch, J. Meyers, K. N. Abazajian, E. Altamura, C. Baccigalupi, K. K. Boddy, T. Brinckmann and Y. Chinone, *et al.*
Sensitivity of Next-Generation CMB Surveys to Neutrinos and Other Light Relics,
arXiv:2608.07453 [astro-ph.CO], Submitted to JCAP.
3. E. Guzman and J. Meyers,
Deep Learning for Primordial *B*-mode Extraction,
Phys. Rev. D **113**, no.6, 063573 (2026) arXiv:2512.19577 [astro-ph.CO].
4. (*) F. Zhang, H. Diao, B. Li, J. Meyers and P. R. Shapiro,
Accurate neural network emulator for primordial light element abundances,
Phys. Rev. D **114**, no.4, 043535 (2026) arXiv:2512.15266 [astro-ph.CO].
5. I. Abril-Cabezas *et al.* [Simons Observatory],
The Simons Observatory: forecasted constraints on primordial gravitational waves with the expanded array of Small Aperture Telescopes,
JCAP **04**, 051 (2026) arXiv:2512.15833 [astro-ph.CO].
6. C. Cook and J. Meyers,
Insights for Early Dark Energy with Big Bang Nucleosynthesis,
Phys. Rev. D **113**, no.4, 043519 (2026) arXiv:2512.11163 [astro-ph.CO].
7. P. W. Graham, D. Green and J. Meyers,
New interpretations of the cosmological preference for a negative neutrino mass,
Phys. Rev. D **113**, no.4, 043514 (2026) arXiv:2508.20999 [astro-ph.CO].
8. A. Leluc, J. Meyers and A. van Engelen,
Time-Dependent Deflection Reconstruction: A New Technique to Search for Gravitational Waves with the Cosmic Microwave Background,
JCAP **01**, 053 (2026) arXiv:2508.10673 [astro-ph.CO].
9. (*) F. Zhang, Y. Luo, B. Li, R. Cao, W. Peng, J. Meyers and P. R. Shapiro,
SageNet: Fast Neural Network Emulation of the Stiff-amplified Gravitational Waves from Inflation,
Astrophys. J. Suppl. **279**, no.2, 44 (2025) arXiv:2504.04054 [astro-ph.CO].
10. B. Li, J. Meyers and P. R. Shapiro,
Multimodality in the Search for New Physics in Pulsar Timing Data and the Case of Kination-amplified Gravitational-wave Background from Inflation,
Astrophys. J. **985**, no.1, 117 (2025) arXiv:2503.18937 [astro-ph.CO].
11. M. Abitbol *et al.* [Simons Observatory],
The Simons Observatory: Science Goals and Forecasts for the Enhanced Large Aperture

Telescope,
JCAP **08**, 034 (2025) arXiv:2503.00636 [astro-ph.IM].

12. V. C. Chan, R. Hložek, J. Meyers and A. van Engelen,
Running the small-correlated-against-large estimator at scale: Applications of small-scale CMB lensing estimators on realistic simulations,
Phys. Rev. D **113**, no.2, 023535 (2026) arXiv:2409.05326 [astro-ph.CO].

13. D. Green and J. Meyers,
The Cosmological Preference for Negative Neutrino Mass,
Phys. Rev. D **111**, no.8, 083507 (2025) arXiv:2407.07878 [astro-ph.CO].

14. N. Craig, D. Green, J. Meyers and S. Rajendran,
No νs is Good News,
JHEP **09**, 097 (2024) arXiv:2405.00836 [astro-ph.CO].

15. J. R. Bond, G. M. Fuller, E. Grohs, J. Meyers and M. J. Wilson,
Cosmic Neutrino Decoupling and its Observable Imprints: Insights from Entropic-Dual Transport,
JCAP **09**, 014 (2024) arXiv:2403.19038 [astro-ph.CO].

16. (*) C. Trendafilova, S. C. Hotinli and J. Meyers,
Improving Constraints on Inflation with CMB Delensing,
JCAP **06**, 017 (2024) arXiv:2312.02954 [astro-ph.CO].

17. J. Ange and J. Meyers,
Improving Constraints on Models Addressing the Hubble Tension with CMB Delensing,
JCAP **10**, 045 (2023) arXiv:2307.01662 [astro-ph.CO].

18. V. C. Chan, R. Hložek, J. Meyers and A. van Engelen,
Small-correlated-against-large estimator for the lensing of the cosmic microwave background,
Phys. Rev. D **109**, no.4, 043527 (2024) arXiv:2302.13350 [astro-ph.CO].

19. (*) J. Ryan, B. Stevenson, C. Trendafilova and J. Meyers,
Beyond Fisher Forecasting for Cosmology,
Phys. Rev. D **107**, no.10, 103506 (2023) arXiv:2211.06534 [astro-ph.CO].

20. (*) Y. Zhu, B. Beringue, S. K. Choi, N. Battaglia, P. D. Meerburg and J. Meyers,
Estimating the Impact of foregrounds on the Future Detection of Rayleigh scattering,
JCAP **09**, 048 (2022) arXiv:2205.04496 [astro-ph.CO].

21. (*) S. C. Hotinli, J. Meyers, C. Trendafilova, D. Green and A. van Engelen,
The Benefits of CMB Delensing,
JCAP **04**, no.04, 020 (2022) arXiv:2111.15036 [astro-ph.CO].

22. D. Green and J. Meyers,
Cosmological Implications of a Neutrino Mass Detection,
arXiv:2111.01096 [astro-ph.CO].

23. (*) T. Namikawa, A. B. Lizancos, N. Robertson, B. D. Sherwin, A. Challinor, D. Alonso, S. Azzoni, C. Baccigalupi, E. Calabrese and J. Carron, *et al.*
The Simons Observatory: Constraining inflationary gravitational waves with multi-tracer B-mode delensing,
Phys. Rev. D **105**, no.2, 023511 (2022) arXiv:2110.09730 [astro-ph.CO].

24. E. Guzman and J. Meyers,
Reconstructing Cosmic Polarization Rotation with ResUNet-CMB,
JCAP **01**, no.01, 030 (2022) arXiv:2109.09715 [astro-ph.CO].

25. M. Aravena *et al.* [CCAT-Prime],
CCAT-prime Collaboration: Science Goals and Forecasts with Prime-Cam on the Fred Young Submillimeter Telescope,
Astrophys. J. Suppl. **264**, no.1, 7 (2023) arXiv:2107.10364 [astro-ph.CO].

26. M. J. Bustamante-Rosell, J. Meyers, N. Pearson, C. Trendafilova and A. Zimmerman,
Gravitational Wave Timing Array,
Phys. Rev. D **105**, no.4, 044005 (2022) arXiv:2107.02788 [gr-qc].

27. P. C. Breysse, S. Foreman, L. C. Keating, J. Meyers and N. Murray,
Mapping the Universe in HD,
Phys. Rev. D **105**, no.8, 083009 (2022), **Editor's Suggestion** arXiv:2104.06422 [astro-ph.CO].

28. E. Guzman and J. Meyers,
Reconstructing Patchy Reionization with Deep Learning,
Phys. Rev. D **104**, no.4, 043529 (2021) arXiv:2101.01214 [astro-ph.CO].

29. (*) N. Pearson, C. Trendafilova and J. Meyers,
Searching for Gravitational Waves with Strongly Lensed Repeating Fast Radio Bursts,
Phys. Rev. D **103**, no.6, 063017 (2021) arXiv:2009.11252 [astro-ph.CO].

30. K. Abazajian *et al.* [CMB-S4 Collaboration],
CMB-S4: Forecasting Constraints on Primordial Gravitational Waves,
Astrophys. J. **926**, no.1, 54 (2022) arXiv:2008.12619 [astro-ph.CO].

31. (*) B. Beringue, P. D. Meerburg, J. Meyers and N. Battaglia,
Cosmology with Rayleigh Scattering of the Cosmic Microwave Background,
JCAP **01**, 060 (2021) arXiv:2008.11688 [astro-ph.CO].

32. S. C. Hotinli, M. C. Johnson and J. Meyers,
Optimal filters for the moving lens effect,
Phys. Rev. D **103**, no.4, 043536 (2021) arXiv:2006.03060 [astro-ph.CO].

33. A. Laguë and J. Meyers,
Prospects and Limitations for Constraining Light Relics with Primordial Abundance Measurements,
Phys. Rev. D **101**, no.4, 043509 (2020) arXiv:1908.05291 [astro-ph.CO].

34. K. Abazajian *et al.* [CMB-S4 Collaboration],
CMB-S4 Science Case, Reference Design, and Project Plan,
arXiv:1907.04473 [astro-ph.IM].

35. (*) S. Hanany *et al.* [NASA PICO Collaboration],
PICO: Probe of Inflation and Cosmic Origins,
arXiv:1902.10541 [astro-ph.IM].

36. (*) S. C. Hotinli *et al.*,
Transverse Velocities with the Moving Lens Effect,
Phys. Rev. Lett. **123**, no. 6, 061301 (2019) arXiv:1812.03167 [astro-ph.CO].

37. (*) S. Foreman, P. D. Meerburg, J. Meyers and A. van Engelen,
Cosmic variance mitigation in measurements of the integrated Sachs-Wolfe effect,
Phys. Rev. D **99**, 083506 (2019) arXiv:1811.00529 [astro-ph.CO].

38. J. Aguirre *et al.* [Simons Observatory Collaboration],
The Simons Observatory: Science goals and forecasts,
JCAP **1902**, 056 (2019) arXiv:1808.07445 [astro-ph.CO].

39. (*) G. J. Stacey *et al.*,
CCAT-prime: Science with an Ultra-widefield Submillimeter Observatory at Cerro Chajnantor,
arXiv:1807.04354 [astro-ph.GA].

40. R. de Putter, O. Doré, J. Gleyzes, D. Green and J. Meyers,
Dark Matter Interactions, Helium, and the CMB,
Phys. Rev. Lett. **122**, no.4, 041301 (2019) arXiv:1805.11616 [astro-ph.CO].

41. D. Green, P. D. Meerburg and J. Meyers,
Aspects of Dark Matter Annihilation in Cosmology,
JCAP **1904**, 025 (2019) arXiv:1804.01055 [astro-ph.CO].

42. (*) S. Foreman, P. D. Meerburg, A. van Engelen and J. Meyers,
Lensing reconstruction from line intensity maps: the impact of gravitational nonlinearity,
JCAP **1807**, no. 07, 046 (2018) arXiv:1803.04975 [astro-ph.CO].

43. (*) S. C. Hotinli, J. Frazer, A. H. Jaffe, J. Meyers, L. C. Price and E. R. M. Tarrant,
Effect of reheating on predictions following multiple-field inflation,
Phys. Rev. D **97**, no. 2, 023511 (2018) arXiv:1710.08913 [astro-ph.CO].

44. (*) J. Meyers, P. D. Meerburg, A. van Engelen and N. Battaglia,
Beyond CMB Cosmic Variance Limits on Reionization with the Polarized SZ effect,
Phys. Rev. D **97**, no. 10, 103505 (2018), **Editor's Suggestion** arXiv:1710.01708 [astro-ph.CO].

45. P. D. Meerburg, J. Meyers and A. van Engelen,
Reconstructing the Primary CMB Dipole,
Phys. Rev. D **96**, no. 8, 083519 (2017) arXiv:1704.00718 [astro-ph.CO].

46. P. D. Meerburg, J. Meyers, K. M. Smith and A. van Engelen,
Reconstructing CMB Fluctuations and the Mean Reionization Optical Depth,
Phys. Rev. D **95**, no. 12, 123538 (2017) arXiv:1701.06992 [astro-ph.CO].

47. (*) C. Sheere, A. van Engelen, P. D. Meerburg and J. Meyers,
Establishing the Origin of CMB B-mode Polarization,
Phys. Rev. D **96**, no. 6, 063508 (2017) arXiv:1610.09365 [astro-ph.CO].

48. K. N. Abazajian *et al.* [CMB-S4 Collaboration],
CMB-S4 Science Book, First Edition,
arXiv:1610.02743 [astro-ph.CO].

49. R. de Putter, O. Doré, D. Green and J. Meyers,
Single-Field Inflation and the Local Ansatz: Distinguishability and Consistency,
Phys. Rev. D **95**, no. 6, 063501 (2017) arXiv:1610.00785 [hep-th].

50. D. Green, J. Meyers and A. van Engelen,
CMB Delensing Beyond the B Modes,
JCAP **1712**, no.12, 005 (2017) arXiv:1609.08143 [astro-ph.CO].

51. (*) P. D. Meerburg, J. Meyers, A. van Engelen and Y. Ali-Haïmoud,
CMB B-Mode Non-Gaussianity,
Phys. Rev. D **93**, 123511 (2016) arXiv:1603.02243 [astro-ph.CO].

52. D. Baumann, D. Green, J. Meyers and B. Wallisch,
Phases of New Physics in the CMB,
JCAP **1601**, 007 (2016) arXiv:1508.06342 [astro-ph.CO].

53. (*) P. D. Meerburg, R. Hložek, B. Hadzhiyska and J. Meyers,
Multiwavelength Constraints on the Inflationary Consistency Relation,
Phys. Rev. D **91**, no. 10, 103505 (2015) arXiv:1502.00302 [astro-ph.CO].

54. M. Alvarez *et al.*,
Testing Inflation with Large Scale Structure: Connecting Hopes with Reality,
arXiv:1412.4671 [astro-ph.CO].

55. V. Britto and J. Meyers,
Monthly Modulation in Dark Matter Direct-Detection Experiments,
JCAP **1511**, 006 (2015) arXiv:1409.2858 [astro-ph.CO].

56. J. Meyers and E. R. M. Tarrant,
Perturbative Reheating After Multiple-Field Inflation: The Impact on Primordial Observables,
Phys. Rev. D **89**, no. 6, 063535 (2014) arXiv:1311.3972 [astro-ph.CO].

57. J. Meyers,
Non-Gaussian Correlations Outside the Horizon in Local Thermal Equilibrium,
arXiv:1212.4438 [astro-ph.CO].

58. J. Meyers and N. Sivanandam,
Adiabaticity and the Fate of Non-Gaussianities: The Trispectrum and Beyond,
Phys. Rev. D **84**, 063522 (2011) arXiv:1104.5238 [astro-ph.CO].

59. J. Meyers and N. Sivanandam,
Non-Gaussianities in Multifield Inflation: Superhorizon Evolution, Adiabaticity, and the Fate of fnl,
Phys. Rev. D **83**, 103517 (2011) arXiv:1011.4934 [astro-ph.CO].

60. W. Fischler and J. Meyers,
Dark Radiation Emerging After Big Bang Nucleosynthesis?,
Phys. Rev. D **83**, 063520 (2011) arXiv:1011.3501 [astro-ph.CO].

Snowmass 2021 Papers

The papers in this section were written for the Snowmass 2021 DPF Community Planning Exercise. Papers in this section do not have alphabetical authors lists. Those which I co-led are indicated by (‡), and those for which I was among the primary authors are indicated by (†).

61. R. X. Adhikari, *et al.*
Report of the Topical Group on Cosmic Probes of Fundamental Physics for Snowmass 2021,
arXiv:2209.11726 [hep-ph].

62. B. Flaugher, V. Miranda, D. J. Schlegel *et al.*
Report of the Topical Group on Dark Energy and Cosmic Acceleration: Complementarity of Probes and New Facilities for Snowmass 2021,
arXiv:2209.08654 [astro-ph.CO].

63. A. Drlica-Wagner, C. Prescod-Weinstein, H. B. Yu *et al.*,
Report of the Topical Group on Cosmic Probes of Dark Matter for Snowmass 2021,
arXiv:2209.08215 [hep-ph].

64. D. Green, J. T. Ruderman, B. R. Safdi, J. Shelton *et al.*,
Snowmass Theory Frontier: Astrophysics and Cosmology,
arXiv:2209.06854 [hep-ph].

65. J. A. Blazek *et al.*
Snowmass2021 Cosmic Frontier White Paper: Enabling Flagship Dark Energy Experiments to Reach their Full Potential,
arXiv:2204.01992 [astro-ph.CO].

66. (†) R. Brito *et al.*
Snowmass2021 Cosmic Frontier White Paper: Probing dark matter with small-scale astrophysical observations,
arXiv:2203.15954 [hep-ph].

67. (†) C. L. Chang, K. M. Huffenberger *et al.*
Snowmass2021 Cosmic Frontier: Cosmic Microwave Background Measurements White Paper,
arXiv:2203.07638 [astro-ph.CO].

68. (†) K. Abazajian *et al.* [CMB-S4 Collaboration],
Snowmass 2021 CMB-S4 White Paper,
arXiv:2203.08024 [astro-ph.CO].

69. (‡)(†) C. Dvorkin, J. Meyers *et al.*
The Physics of Light Relics,
arXiv:2203.07943 [hep-ph].

70. (†) C. Dvorkin, R. Hlozek, *et al.*
Dark Matter Physics from the CMB-S4 Experiment,
arXiv:2203.07064 [hep-ph].

71. (†) K. N. Abazajian *et al.*
Synergy between cosmological and laboratory searches in neutrino physics: a white

paper,
Phys. Dark Univ. **42**, 101333 (2023) arXiv:2203.07377 [hep-ph].

72. (†) E. J. Baxter, C. Chang, A. Hearin *et al.*
Snowmass2021: Opportunities from Cross-survey Analyses of Static Probes,
arXiv:2203.06795 [hep-ex].

73. S. Aiola *et al.* [CMB-HD],
Snowmass2021 CMB-HD White Paper,
arXiv:2203.05728 [astro-ph.CO].

Astro2020 Decadal Survey White Papers

The papers in this section were written for the Astro2020 Decadal Survey of Astronomy and Astrophysics. Papers in this section do not have alphabetical authors lists, and those for which I was among the primary authors are indicated by (†).

74. N. Sehgal, *et al.*,
CMB-HD: Astro2020 RFI Response,
arXiv:2002.12714 [astro-ph.CO].

75. T. Herter *et al.*,
The CCAT-Prime Submillimeter Observatory,
Bull. Am. Astron. Soc. **51**, no. 7, 213 (2019) arXiv:1909.02587 [astro-ph.IM].

76. S. Hanany *et al.*,
PICO: Probe of Inflation and Cosmic Origins,
Bull. Am. Astron. Soc. **51**, no. 7, 194 (2019) arXiv:1908.07495 [astro-ph.IM].

77. J. Carlstrom *et al.*,
CMB-S4,
Bull. Am. Astron. Soc. **51**, no. 7, 209 (2019) arXiv:1908.01062 [astro-ph.IM].

78. A. Lee *et al.* [Simons Observatory Collaboration],
The Simons Observatory,
Bull. Am. Astron. Soc. **51**, no. 7, 147 (2019) arXiv:1907.08284 [astro-ph.IM].

79. N. Sehgal *et al.*,
CMB-HD: An Ultra-Deep, High-Resolution Millimeter-Wave Survey Over Half the Sky,
Bull. Am. Astron. Soc. **51**, no. 7, 6 (2019) arXiv:1906.10134 [astro-ph.CO].

80. A. Slosar *et al.*,
Dark Energy and Modified Gravity,
arXiv:1903.12016 [astro-ph.CO].

81. A. Slosar *et al.*,
Scratches from the Past: Inflationary Archaeology through Features in the Power Spectrum of Primordial Fluctuations,
Bull. Am. Astron. Soc. **51**, no. 3, 98 (2019) arXiv:1903.09883 [astro-ph.CO].

82. (†) E. B. Grohs, J. R. Bond, R. J. Cooke, G. M. Fuller, J. Meyers and M. W. Paris,
Big Bang Nucleosynthesis and Neutrino Cosmology,
Bull. Am. Astron. Soc. **51**, no. 3, 412 (2019) arXiv:1903.09187 [astro-ph.CO].

83. S. Ferraro *et al.*,
Inflation and Dark Energy from Spectroscopy at $z > 2$,
Bull. Am. Astron. Soc. **51**, no. 3, 72 (2019) arXiv:1903.09208 [astro-ph.CO].

84. (†) D. Green *et al.*,
Messengers from the Early Universe: Cosmic Neutrinos and Other Light Relics,
Bull. Am. Astron. Soc. **51**, no. 3, 159 (2019) arXiv:1903.04763 [astro-ph.CO].

85. S. Shandera *et al.*,
Probing the origin of our Universe through cosmic microwave background constraints on gravitational waves,
Bull. Am. Astron. Soc. **51**, no. 3, 338 (2019) arXiv:1903.04700 [astro-ph.CO].

86. P. D. Meerburg *et al.*,
Primordial Non-Gaussianity,
Bull. Am. Astron. Soc. **51**, no. 3, 107 (2019) arXiv:1903.04409 [astro-ph.CO].

87. N. Sehgal *et al.*,
Science from an Ultra-Deep, High-Resolution Millimeter-Wave Survey,
Bull. Am. Astron. Soc. **51**, no. 3, 43 (2019) arXiv:1903.03263 [astro-ph.CO].

Conference Proceedings

88. J. Meyers,
Cosmic Neutrinos and Other Light Relics,
arXiv:1605.05575 [astro-ph.CO].